Cross-Dataset Evaluation of mBERT and MuRIL for Hinglish Hate and Offensive Speech Detection

Om Patnaik | Working Paper Draft v0.2 - August 25, 2026

Abstract

Hinglish hate speech detection is difficult because online text often mixes Hindi and English, uses inconsistent Romanization, moves between Latin and Devanagari scripts, and changes sharply across platform, topic, and annotation policy. This project compares mBERT, a general multilingual BERT model, with MuRIL, an Indian-language-focused model, for binary hate/offensive speech detection across multiple Hindi-English code-mixed dataset situations. Current experiments show that neither model is universally superior. mBERT performs better on matched Latin-script Hinglish and code-mixed offensive datasets, while MuRIL performs better on targeted religious hate and several THAR-related transfer settings. Cross-dataset evaluation reveals substantial generalization gaps and shows that TF-IDF baselines remain competitive in some transfer conditions.

1. Research Motivation and Claim

The original research question asked whether MuRIL, because it was designed for Indian languages, would outperform mBERT for Hinglish hate speech detection. The experiments so far show that the answer cannot be reduced to one universal winner. The stronger research claim is conditional: model ranking changes with dataset content, platform, script composition, and positive-label definition.

- mBERT is stronger on matched Kaggle Hinglish hate and CM code-mixed offensive evaluation.
- MuRIL is stronger on matched THAR targeted religious hate and on several THAR-related transfer settings.
- Every primary test dataset is best served by an in-domain model, showing that cross-dataset robustness is limited.
- TF-IDF baselines are competitive in several settings, so transformer performance must be interpreted against lexical baselines.

2. Related Work

This project sits at the intersection of multilingual language models, Indian-language representation learning, and code-mixed harmful speech detection. mBERT is a general multilingual BERT model, while MuRIL was introduced to improve representation learning for Indian languages, including translated and transliterated signals. This makes MuRIL a natural candidate for Hinglish and Hindi-English code-mixed social media text.

Prior Hindi-English code-mixed hate speech work, including the Bohra et al. dataset, shows that this task cannot be treated as ordinary English hate speech detection. Code-mixed text contains unstable spelling, transliteration, language switching, and local social context. THAR adds another important angle by focusing specifically on targeted religious hate in Hindi-English code-mixed comments. The current project builds on these ideas but places emphasis on cross-dataset robustness: whether a model trained on one dataset situation still works when the platform, target, script mix, or positive-label definition changes.

3. Datasets and Dataset Situations

The project uses three primary source-backed datasets. The 79-row benchmark is excluded from primary claims because its provenance is uncertain and it may be manually written or AI-generated.

Dataset	Source/domain	Rows	Label 0	Label 1	Positive meaning
Kaggle Hinglish	Kaggle Hinglish subset	4,780	2,914	1,866	Hate
CM code-mixed	Indian politics Twitter/X	3,900	2,455	1,445	Offensive
THAR religion	YouTube religion comments	11,549	6,095	5,454	AntiReligion

Kaggle Hinglish is fully Latin-script in the processed subset and is useful for Romanized Hinglish behavior. CM is mostly Latin-script but contains Indian political content, many mentions, hashtags, and an offensive rather than strict hate label. THAR is narrower but highly relevant: it focuses on targeted religious hate in Hindi-English code-mixed YouTube comments and contains more Devanagari and mixed-script text.

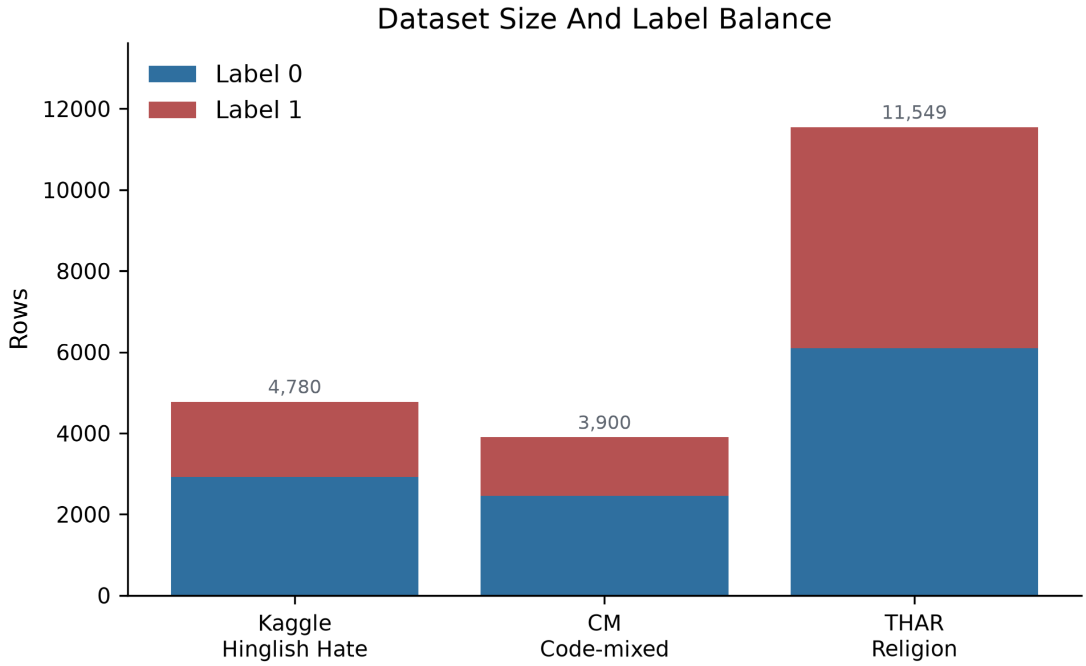

Figure 1. Dataset size and label balance.

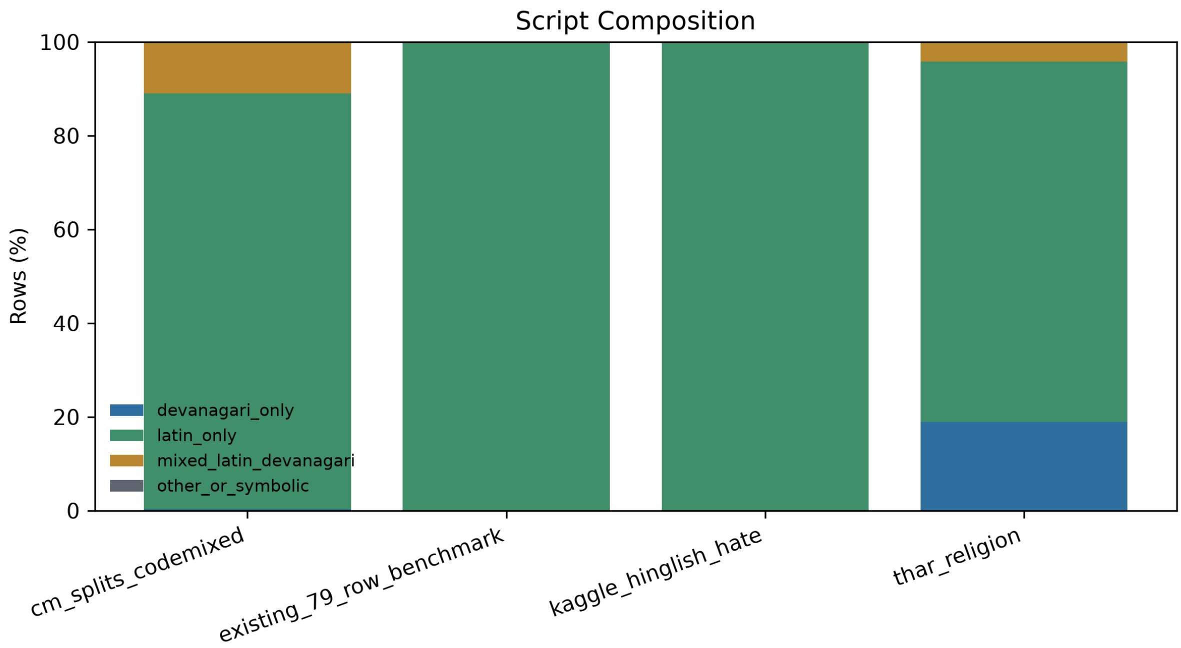

Figure 2. Script composition by dataset.

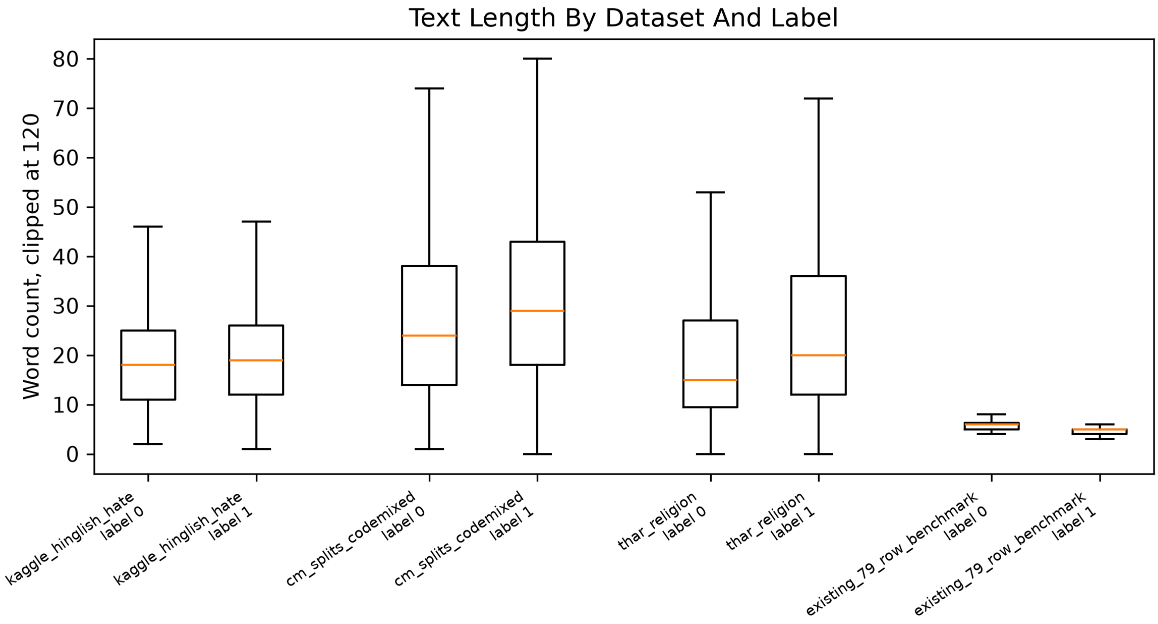

Figure 3. Text length by label.

4. Models and Experimental Setup

The transformer experiments compare bert-base-multilingual-cased for mBERT and google/muril-base-cased for MuRIL. Each model is fine-tuned as a binary sequence classifier. The controlled checkpoints use seed 42, two training epochs, maximum sequence length 128, learning rate 2e-5, and minimal URL/user normalization. For datasets without an official split, a stratified 80/20 split is used. For CM, source-provided splits are used.

The project also evaluates TF-IDF with logistic regression and TF-IDF with linear SVM. These baselines are important because hate and offensive datasets can contain strong repeated lexical cues. A transformer claim is only meaningful if it is read alongside these simpler models.

Reproducibility Box

Item	Current setting
Models	bert-base-multilingual-cased; google/muril-base-cased
Primary datasets	Kaggle Hinglish; CM code-mixed; THAR religion
Split policy	Stratified 80/20 where needed; CM source split used
Seed	42 for current controlled transformer runs
Epochs	2
Max sequence length	128
Learning rate	2e-5
Baselines	TF-IDF + logistic regression; TF-IDF + linear SVM
Hardware/scripts	Device-specific wrappers for Mac MPS, CPU/debug, and CUDA/Colab
Known rigor gap	Multi-seed runs and confidence intervals are still pending

Two epochs were chosen as a controlled first-pass setting so every model/dataset condition could be compared under the same budget. This is not a claim that two epochs are optimal. Future versions should use multi-seed runs, validation curves, and possibly early stopping.

5. Primary Matched Results

Matched evaluation means that a model is trained and tested within the same dataset situation. This is the cleanest setting for measuring how well each pretrained model adapts to a specific dataset.

Test dataset	Model	Acc.	Pos. recall	Pos. F1	Macro F1	TN/FP/FN/TP
Kaggle Hinglish	mBERT	71.4%	38.6%	51.3%	65.6%	539/44/229/144
Kaggle Hinglish	MURIL	67.5%	22.3%	34.8%	56.6%	562/21/290/83
CM code-mixed	mBERT	80.5%	68.7%	71.4%	78.3%	233/35/46/101
CM code-mixed	MURIL	78.8%	55.1%	64.8%	74.8%	246/22/66/81
THAR religion	mBERT	74.8%	77.7%	74.5%	74.8%	880/339/243/848
THAR religion	MURIL	77.9%	80.3%	77.4%	77.9%	923/296/215/876

Interpretation: mBERT wins the matched Kaggle and CM conditions, while MuRIL wins the matched THAR condition. This is the first major reason the paper should avoid claiming a universal winner.

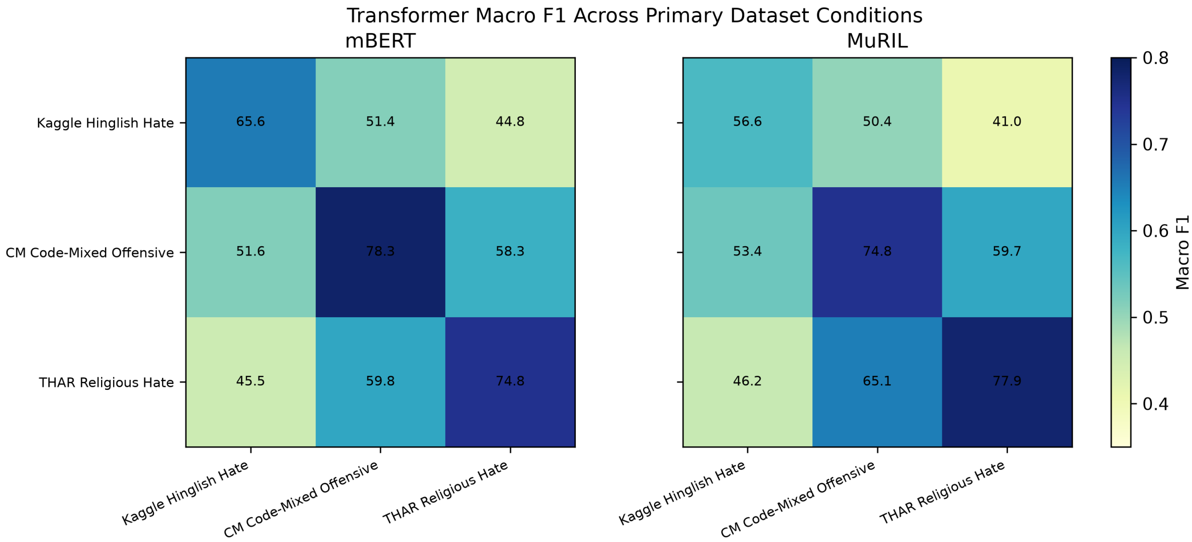

Figure 4. Primary transformer macro F1 matrix.

6. Cross-Dataset Transfer

Cross-dataset evaluation asks whether a model trained on one dataset can generalize to another. This is the main research contribution because it exposes whether a model has learned broadly useful hate/offense signals or only the local annotation and lexical patterns of one dataset.

Train	Test	Macro winner	Macro gap	Recall winner	Recall gap
Kaggle Hinglish	Kaggle Hinglish	mBERT	9.0%	mBERT	16.4%
CM code-mixed	CM code-mixed	mBERT	3.5%	mBERT	13.6%
THAR religion	THAR religion	MURIL	3.1%	MURIL	2.6%
CM code-mixed	THAR religion	MURIL	1.4%	MURIL	14.8%
THAR religion	CM code-mixed	MURIL	5.3%	MURIL	12.9%
Kaggle Hinglish	THAR religion	mBERT	3.8%	mBERT	7.6%
THAR religion	Kaggle Hinglish	MURIL	0.6%	MURIL	2.7%

MuRIL becomes stronger in THAR-related transfer, especially CM-to-THAR and THAR-to-CM. However, both models transfer poorly between broad Hinglish hate and targeted religious hate. Kaggle-trained models are very conservative on THAR, while THAR-trained models do not become general Hinglish hate detectors.

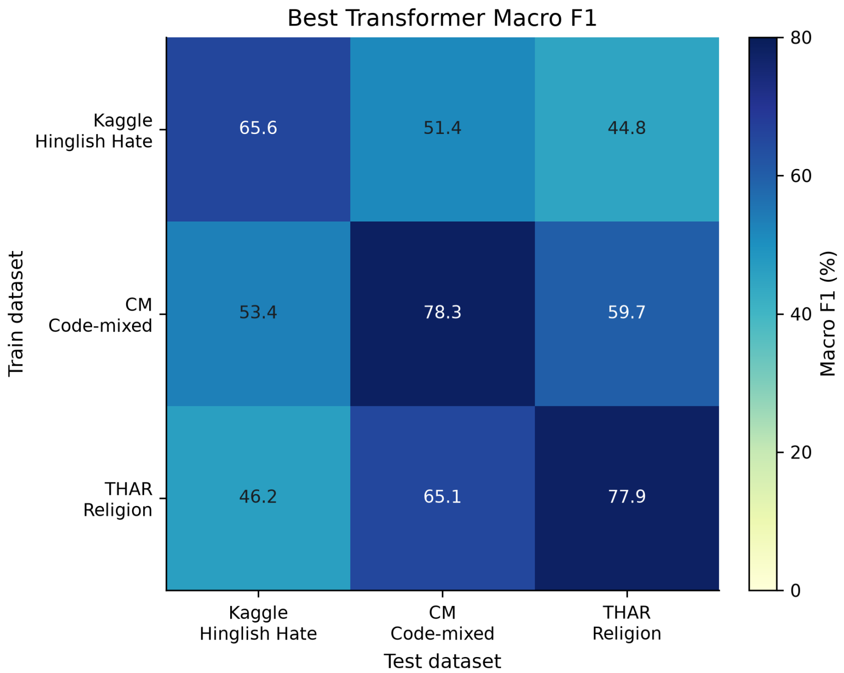

Figure 5. Transformer cross-dataset macro F1.

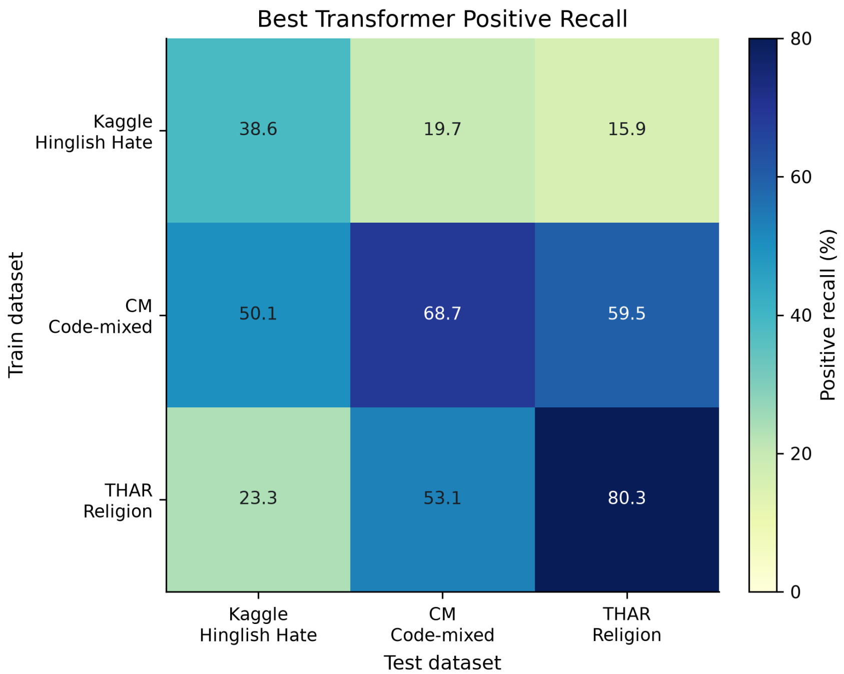

Figure 6. Transformer cross-dataset positive recall.

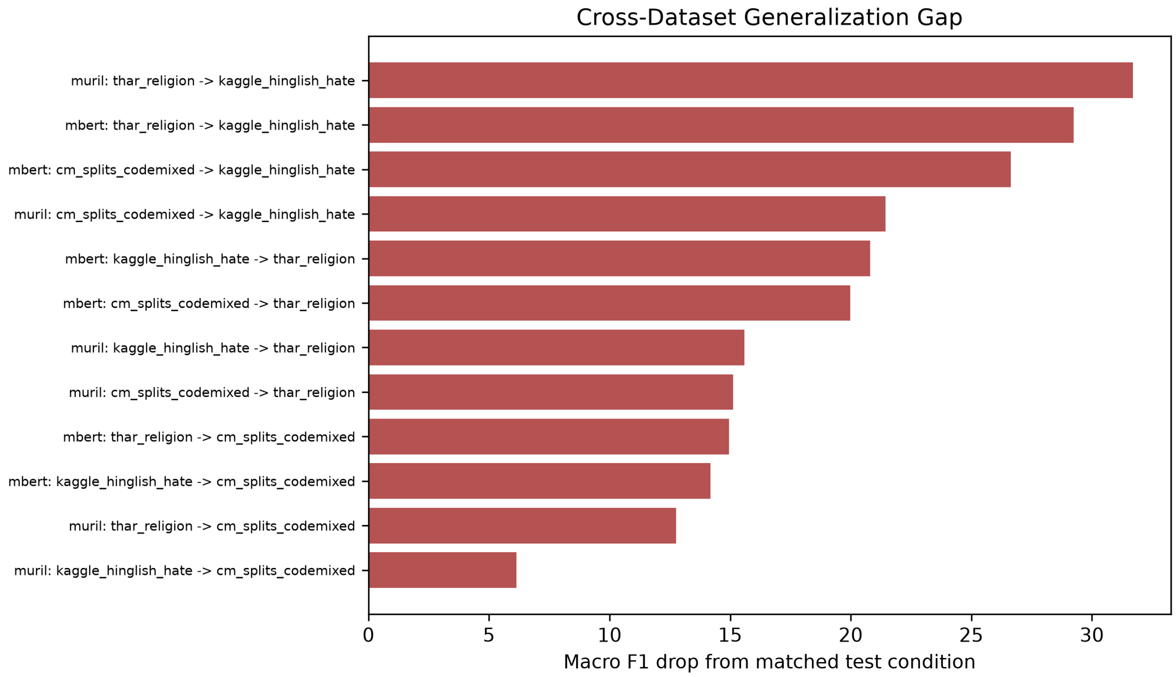

Figure 7. Generalization gaps from matched to transfer evaluation.

7. Baselines and Lexical Effects

TF-IDF baselines are not the focus of the project, but they are essential controls. If TF-IDF performs well, that suggests strong lexical regularities in the dataset. In several transfer settings, TF-IDF beats the best transformer, especially Kaggle-to-THAR and THAR-to-Kaggle.

Train	Test	Best transformer	Trans. macro	Best TF-IDF	TF-IDF macro	Delta
CM code-mixed	CM code-mixed	mBERT	78.3%	Logistic Regression	77.6%	0.7%
Kaggle Hinglish	Kaggle Hinglish	mBERT	65.6%	Logistic Regression	61.4%	4.1%
THAR religion	THAR religion	MURIL	77.9%	Logistic Regression	70.6%	7.3%
Kaggle Hinglish	THAR religion	mBERT	44.8%	Logistic Regression	49.9%	-5.2%
THAR religion	Kaggle Hinglish	MURIL	46.2%	Logistic Regression	51.0%	-4.9%
CM code-mixed	THAR religion	MURIL	59.7%	Linear Svm	53.8%	5.9%
THAR religion	CM code-mixed	MURIL	65.1%	Logistic Regression	60.9%	4.2%

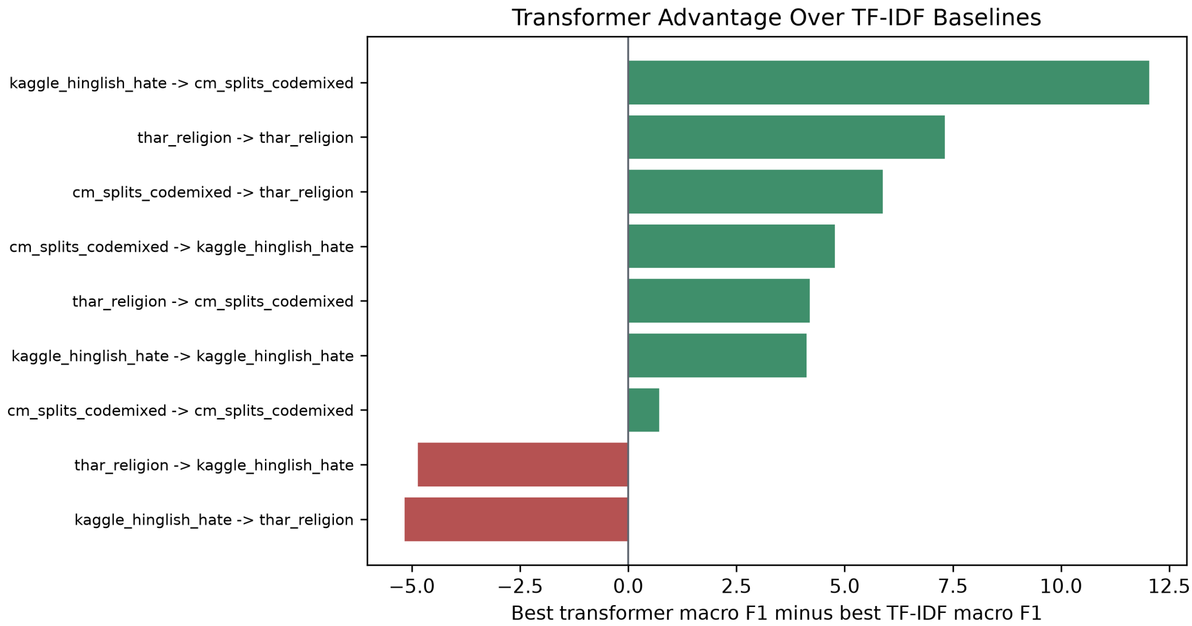

Figure 8. Transformer macro F1 minus best TF-IDF macro F1.

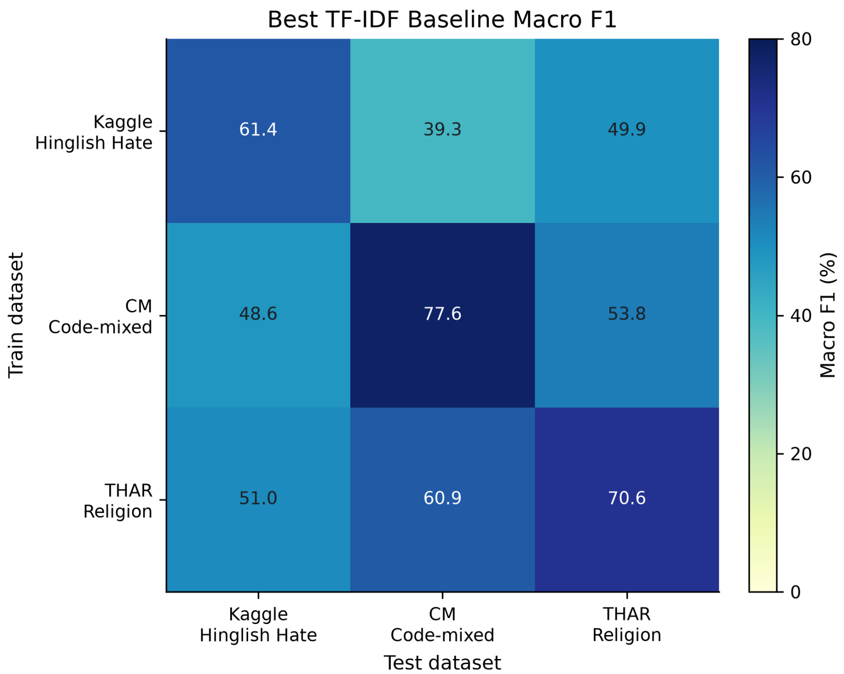

Figure 9. Baseline cross-dataset macro F1.

8. Error Analysis

False negatives are especially important in this task because they are harmful examples predicted as non-harmful. The false-negative pattern matches the larger story: mBERT misses fewer positives in matched Kaggle and CM, while MuRIL misses fewer positives in matched THAR and THAR-to-CM transfer.

Train	Test	Rows	Pos.	mBERT FN	MuRIL FN	mBERT FP	MuRIL FP
Kaggle Hinglish	Kaggle Hinglish	956	373	61.4%	77.7%	7.5%	3.6%
CM code-mixed	CM code-mixed	415	147	31.3%	44.9%	13.1%	8.2%
THAR religion	THAR religion	2310	1091	22.3%	19.7%	27.8%	24.3%
Kaggle Hinglish	THAR religion	2310	1091	84.1%	91.8%	14.6%	6.8%
THAR religion	Kaggle Hinglish	956	373	79.4%	76.7%	26.4%	28.8%
THAR religion	CM code-mixed	415	147	59.9%	46.9%	20.9%	23.1%

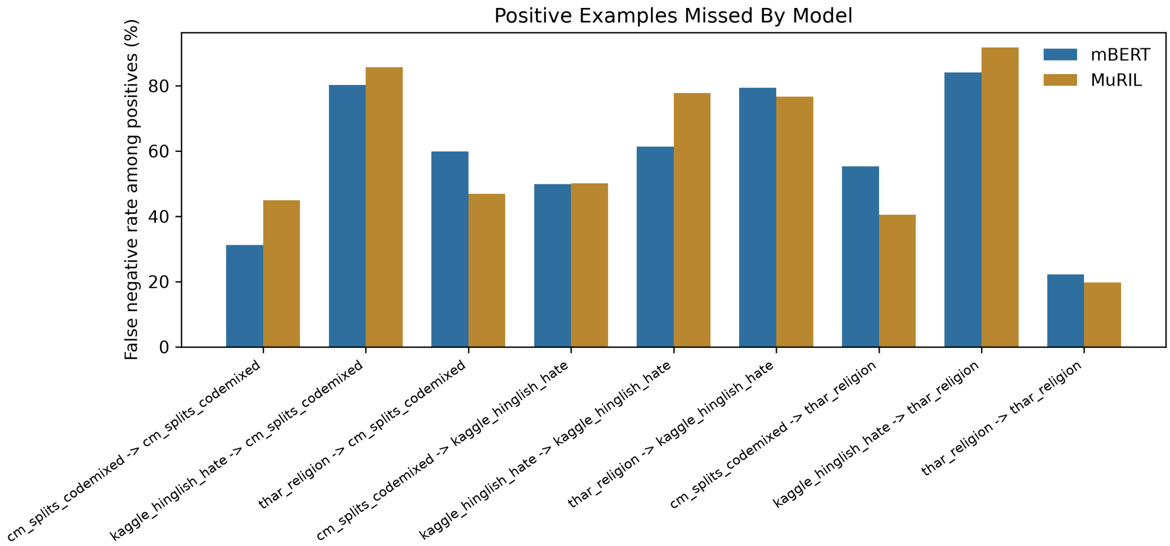

Figure 10. False-negative rates by train/test condition.

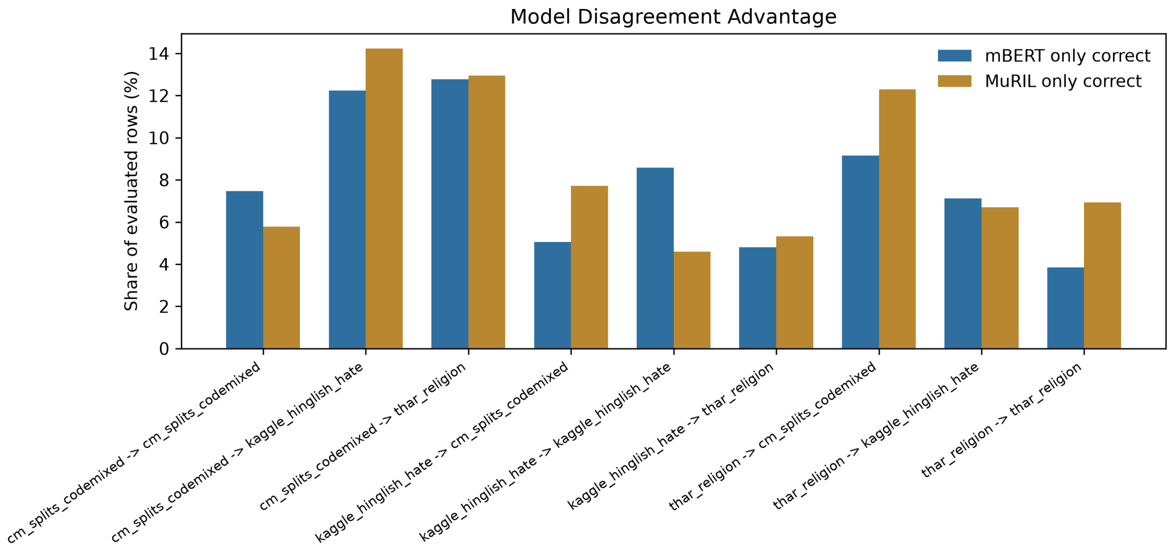

Figure 11. Model disagreement advantage analysis.

Manual Error Coding

A first-pass manual coding of sampled errors was created as a structured reading aid, not final human-ground-truth annotation. The dominant theme is cross-dataset label mismatch: many errors occur because one dataset's positive class is not the same as another dataset's positive class.

Manual reason	Rows	Share
cross dataset label mismatch	191	67.0%
generic profanity or abuse	78	27.4%
target group or religion cue	64	22.5%
political context or slogan	47	16.5%
implicit context or unclear signal	37	13.0%
non hateful lexical trigger	27	9.5%
short or contextless text	26	9.1%
mixed script complexity	9	3.2%
script mismatch devanagari	8	2.8%

9. Main Inferences

- The project should claim conditional model behavior, not absolute model superiority.
- mBERT's strengths appear in mostly Latin-script Hinglish and offensive/code-mixed settings.
- MuRIL's strengths appear in targeted religious hate and some Indian-context transfer settings.
- Dataset label meaning is a major confound: hate, offense, and AntiReligion are related but not interchangeable.
- Cross-dataset failure is not noise; it is a core research result showing weak robustness across dataset situations.
- TF-IDF competitiveness implies that lexical shortcuts and dataset-specific words are part of the story.

10. Limitations

- The three primary datasets differ in platform, label policy, target domain, and script mix.
- CM should be described as offensive/hate-adjacent unless its label definition is verified further.
- The 79-row benchmark is excluded from primary conclusions because its provenance is uncertain.
- The THAR citation and CM licensing/citation details need final verification before formal submission.
- Indo-HateSpeech was reviewed but not added to primary experiments because of short texts, duplicates, source concentration, and label ambiguity.
- Mixed-dataset training has not yet been completed in this draft.

11. Next Experimental Step

The next stage is mixed-dataset training. The project should train mBERT and MuRIL on Kaggle+CM, Kaggle+THAR, CM+THAR, and Kaggle+CM+THAR, then evaluate each checkpoint separately on every primary dataset. This will show whether broader training improves robustness or simply mixes incompatible label definitions.

AI-Use Statement

AI tools were used for coding assistance, environment debugging, documentation organization, and draft generation. The research direction, dataset selection decisions, experiment interpretation, and manual error-analysis decisions were reviewed and directed by the student.

References

- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT.
- Khanuja, S. et al. (2021). MuRIL: Multilingual Representations for Indian Languages. arXiv:2103.10730.
- Bohra, A., Vijay, D., Singh, V., Akhtar, S. S., and Shrivastava, M. (2018). A Dataset of Hindi-English Code-Mixed Social Media Text for Hate Speech Detection. PEOPLES.
- Dhekane, S. Code-Mixed Hinglish Hate Speech Detection Dataset. Kaggle. Accessed 2026-07-01.
- cm-hate-speech-detection contributors. GitHub repository: https://github.com/shikharras/cm-hate-speech-detection. Accessed 2026-07-01.
- Sharma, D. et al. (2024). THAR: Targeted Hate Speech Against Religion. DOI: 10.1145/3653017.